Concept2Cure.RI

TECHNICAL WHITE PAPER

One governed chain from evidence to filing

An AI-native, 21 CFR Part 11-governed regulatory platform spanning medical device and drug pathways — from grant funding and protocol design, through grounded authoring and validated agency transmission, to post-approval change control.

931	359	107	19,181
TABLES FROM A BLANK DATABASE	REGULATORY AI TOOLS	PRODUCT SURFACES	TESTS PASSING

Concept2Cure, Inc.
Prepared for prospective investors · August 2026
Assessed against the `concept2cure-v2` engineering branch

Alongside its capabilities, this paper reports the severe findings of an independent readiness audit and the verified state of each one in the current code. Figures are derived directly from the repository; where a figure is drawn from that audit it is dated, because the audit was a snapshot and the platform has moved since. Forward-looking statements regarding roadmap, market conditions, and competitive position are estimates, not guarantees of outcome.

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A NOTE ON METHOD

Every quantitative claim here is derived from the repository as it stands today. Figures drawn from the August audit are dated as such: that audit was a snapshot, substantial work has landed since, and its aggregate scores are reported as history rather than current state. A vendor selling into regulated life sciences is bought or rejected on whether its claims survive scrutiny, so these are written to be checked.

SECTION 01

Executive summary

Regulatory submission software is fragmented into four silos that no vendor spans. Concept2Cure.RI is built as the spanning layer — and it has been tested against an outside audit, which is the most useful evidence available about how this team builds.

The market failure

Bringing a medical product to market requires assembling evidence, drafting an argument from it, packaging that argument to an agency's exact technical specification, and transmitting it through a validated channel — with every step attributable and defensible years later. That work is split across four categories of vendor that do not talk to each other: evidence and literature platforms, AI authoring startups, publishing and validation incumbents, and regulatory information management systems. Sponsors buy three or four and stitch the seams with consultants.

What has been built

A single governed system spanning that chain — and, it turns out, considerably more than the chain. Source documents enter a tenant-isolated evidence store; a retrieval-grounded assistant drafts against them with per-claim citations; the editor refuses to save content lacking clause-level provenance; approval runs through an electronic-signature workflow; and the package is assembled to eCTD or eSTAR structure, validated, and handed to one of thirteen agency transport adapters. Either side of that sit grant management, protocol development, ethics submissions, risk-based monitoring, clinical study reports, quality and QC, labeling, and post-approval change control. On a blank PostgreSQL the installer builds **931 tables and 787 row-level security policies** and exits clean.

The audit, and what it produced

A 15-domain independent readiness audit, executed against a live instance on 10 August 2026, returned not ready for general availability. Its central observation was not absent features but *controls that reported success while doing nothing* — an integrity monitor verifying an empty chain as intact, RLS policies rendered inert by a superuser connection, an eCTD path emitting zero leaf files. In the four days that followed, **149 commits** landed as a direct remediation programme. **Fourteen of the severe mechanisms it named were re-checked in the code for this paper and found closed**, including every one above. Section 10 gives the finding-by-finding state.

WHY THAT SEQUENCE IS IN AN INVESTOR DOCUMENT

Because it is the most informative thing available about this company. The audit establishes that the architecture is worth keeping — the auditor's own words were "not a facade… a genuinely engineered spine" — and the response establishes how fast this team closes a named list. An investor's real question about a pre-GA platform is not whether defects exist. It is whether the team finds them, reports them, and fixes them at a measurable rate. Here that rate is on the record.

The honest position

The platform remains pre-GA. Running production under enforced tenant isolation, consolidating the audit substrate, and raising request-path coverage are the live engineering items; a first filing additionally depends on licensed agency templates, validator licences, gateway credentials, and a qualification signed by a competent person. The audit's estimate — a caveated design-partner pilot in weeks, a defensible first paying pharma customer in months — is the working plan, and Section 12 sets it out. Its aggregate scores are not restated here as current, because they were a snapshot and no second audit has re-measured them.

SECTION 02

The problem and the market window

The submission market is not underserved — it is mis-served, by vendors each solving one quarter of a problem that only makes sense as a whole.

The work, and where it breaks

A regulatory submission is an argument built from evidence, formatted to a specification, and defended under inspection. Four things must hold simultaneously: the evidence must be found and appraised; the argument must be written and cited; the package must conform byte-for-byte to an agency's technical standard; and every step must be attributable to a person and a moment. No single vendor delivers all four, so the handoffs between them are where sponsors lose months — re-keying content between systems, reconciling citations by hand, and discovering conformance failures only at the gateway.

The timing: a mandatory migration reopens every stack

eSTAR v7.0 — mandatory since 3 Aug 2026

Every 510(k) and De Novo now flows through eSTAR. FDA technical-screening error rates fell from 20.5% to 5.4% under it. The *formatting* pain is shrinking; value is migrating to predicate strategy, testing rationale, and clinical evidence — the content problem.

eCTD 4.0 — the 2026–2029 window

PMDA mandatory April 2026, EMA around 2027, FDA around 2028–29. Every sponsor must migrate. Only about four vendors passed the EMA v4 pilot. This is the decade's one industry-wide drop in switching costs.

Gateway access is open

FDA's ESG NextGen retired the legacy WebTrader client in April 2025 in favour of a portal, REST API, and AS2. There is **no FDA certification programme for eCTD software** — validation criteria are published and a new entrant can offer direct submission on sponsor-held accounts after a short test phase.

AI is being normalized — and governed

FDA deployed its own generative system in 2025; FDA and EMA published joint AI guiding principles in January 2026; EU AI Act obligations phase in from August 2026. AI content is becoming acceptable, which turns *governance of the AI* into the procurement question.

What that combination means

A buyer who must migrate to a new submission standard anyway, who is now permitted to use AI but must prove how it is controlled, and who currently pays three vendors to cover one workflow, is a buyer whose stack is genuinely re-openable. That condition does not persist. It closes when the migration completes and the incumbents' AI bolt-ons mature — which is the clock this business is running against, and the reason the remediation programme in Section 12 is sequenced for speed rather than elegance.

THE ONE THING THAT DOES NOT CHANGE

Whatever the standard or the model generation, a submission must be defensible: an inspector, years later, must be able to ask where a sentence came from and receive an answer. That requirement is indifferent to AI progress, and it is the requirement the platform's architecture is organized around. Frontier capability will keep improving for every competitor simultaneously; the ability to prove provenance will not arrive by upgrading a model.

Market facts are drawn from a competitive benchmark synthesized from public agency notices and industry research in August 2026, with per-claim sources retained in the platform's engineering documentation. Vendor capability descriptions reflect publicly stated positioning at that date; pricing is directional and is not reproduced here.

SECTION 03

The competitive landscape

Four silos, on both the device and the drug side, with the same structural boundary in each. The gap is not a niche — it is the middle of the workflow.

SILO	REPRESENTATIVE VENDORS	WHERE IT STOPS
Evidence & literature	Celegence CAPTIS, DistillerSR, CiteMed, Nested Knowledge, Basil Systems	Produces screened evidence and landscape data, not a submission-shaped argument.
AI authoring	Weave Bio, Peer AI, Yseop, Narrativa, Complizen, Cruxi, Essenvia	Stops at the document boundary — no assembly, validation, or transport.
Publishing & validation	Lorenz docuBridge, Extedo, Certara GlobalSubmit, Ennov	Validates and transmits; AI is a workflow bolt-on, not a drafting engine.
RIM & lifecycle	Veeva Vault RIM, Rimsys, RegDesk, Greenlight Guru	Tracks registrations, process, and quality; does not generate the content.

The notable positions

Veeva Vault RIM — the gravity well

Market leader with several hundred RIM organizations and a downmarket SKU. The durable advantage is distribution, not capability: recurring complaints concern speed at scale, a cumbersome data model, and services overhead, and its 2026 AI is workflow agents rather than drafting. This is the incumbent whose response determines the size of the window.

Lorenz — the validation reference

docuBridge plus eValidator, the industry's reference validation tool, and an FDA eCTD 4.0 pilot partner since 2022. Criticised on desktop-era usability. Relevant here as both a benchmark and a dependency: the platform carries an adapter seam for eValidator, and the licence is a procurement item.

Weave Bio — the closest bridge

AI IND drafting extended to NDA workflow in April 2026 via a contract-research channel, with a published sponsor case reporting a large time reduction and no critical errors. It has no assembly, validation, or gateway — the bridge is a services partnership, not a product.

Greenlight Guru — the governance badge

Medtech quality-system incumbent, ISO/IEC 42001 certified — first to make AI governance a marketable credential in this segment. It does not generate or fill eSTAR. Its certification is the clearest signal that *how the AI is governed* has become a purchasing criterion.

The structural finding

Nobody spans intake → evidence → drafting → official package → post-market. The two closest moves are recent and partial: a RIM vendor added authoring-lite in May 2026 with no evidence engine behind it, and an AI authoring startup reached the drug workflow through a CRO channel. Meanwhile a publishing incumbent has stated publicly that it is evaluating the future of its regulatory writing unit — a signal that the services-attached model is under pressure from exactly this direction.

THE HONEST COUNTERWEIGHT

Every vendor above has something this platform does not: paying customers, executed submissions, and a validation history. Architecture spanning the chain is a real advantage only once it is trustworthy enough to be bought, which is what Sections 10 through 12 are about. The competitive claim in this paper is about product shape, not market position.

SECTION 04

Platform architecture

A single TypeScript monorepo organized in five layers, each of which the layer above can only reach through a governed interface.

Surface LAYER 5	React 18 + Vite single-page application. **Chat-first**: capability is invoked by asking the assistant, not by navigating to a screen per feature. The canonical workspace is an asymmetric three-zone grid — contextual tree, an intelligence stream, and a live-rendered artifact pane that never inherits chat state. A registry of 28 governed components is the single source of truth; raw form elements are rejected by a CI gate.
Governance LAYER 4	The compliance spine, described in Section 06. Audit chaining, electronic signature with signing-authority resolution and version binding, document freeze and lock, clause-level lineage enforced in-transaction, and a default-deny authorization boundary mounted once ahead of all route registration — so no route file can forget authentication.
Intelligence LAYER 3	AnA, the tool-calling regulatory assistant (**359 registered tools** across 13 capability categories); the AI gateway that every model call must pass through; and the RIM, a deterministic non-LLM layer that accumulates regulatory patterns over time. Detailed in Section 05.
Domain services LAYER 2	**217 service domains** covering the regulatory subject matter: eCTD assembly and lifecycle operations, eSTAR generation, substantial-equivalence and predicate analysis, clinical evaluation and literature appraisal, biostatistics, CMC, pharmacovigilance, quality management, and thirteen agency transport gateways. Exposed through **3,846 route handlers** in 389 modules.
Substrate LAYER 1	PostgreSQL with pgvector, addressed through Drizzle. A fresh-database installer provisions **931 tables and 787 row-level security policies** from blank in one idempotent command, takes a cluster-wide lock, re-verifies its readiness contract, and fails closed with a distinct exit code. Object storage for source documents, Redis for cache and queueing, and a real-time collaboration backend for the editor.

Design decisions that matter commercially

Fail-closed by default

The server refuses to boot without its secrets and names what is missing. The AI gateway refuses to serve with no provider. The readiness endpoint fails closed on an incomplete schema. A regulated buyer's first question is what the system does when it is wrong.

One of everything

An enforced doctrine that there is never more than one implementation of a capability, backed by CI gates rejecting duplicate table definitions, unreachable migrations, and parallel runtime paths. It is why a system of this size stays modifiable.

Refusal as a first-class result

When the platform lacks a licensed template, a dataset, or a credential, it says so explicitly rather than approximating. End-to-end journey tests assert honest refusals as passing outcomes. A confident wrong answer is a liability here, not a feature.

WHY A MONOREPO OF THIS DEPTH IS THE ASSET

Regulatory subject matter does not compress. eCTD backbone construction, lifecycle operations, controlled vocabularies, region-specific validation profiles, substantial-equivalence logic, and thirteen distinct agency transport protocols are each irreducibly detailed, and each must interoperate with the governance layer to be usable. The volume of domain code is not incidental scale — it is the barrier a point-solution competitor would have to cross to span the same chain.

SECTION 05

The intelligence layer

Three components: an assistant that acts through governed tools, a gateway that no model call may bypass, and a deterministic layer that learns without a model at all.

AnA — the regulatory assistant

AnA is the product surface. Rather than a chatbot bolted beside a feature set, it is the invocation mechanism for the feature set: **359 tools** spanning submission and eCTD, evidence and intelligence, device and IVD, authoring, clinical, governance, quality management, CMC, nonclinical, biostatistics, and clinical pharmacology. Context is assembled from three memory layers before each turn — the working thread, project memory retrieved by vector similarity, and account context.

> **Grounding is checked deterministically**
>
> After each turn, a self-verification round — no additional model call — extracts the verifiable identifiers from the answer (trial registrations, PMIDs and DOIs, FDA submission numbers) and any quoted source text, and checks each against the evidence the assistant actually gathered that turn. Claims absent from that corpus are surfaced as unsupported. **The verdict is advisory: it is shown to the user, not used to block or rewrite the answer.** Binding retrieval-verified citations into the document itself is Phase 2 work in Section 12.

The AI gateway — a policy chokepoint

Every model call passes through one gateway, and a CI gate rejects code that attempts to bypass it. The gateway performs provider routing and failover, health tracking, cost accounting, and two evaluation passes before dispatch: a synchronous check covering token budget, role-aware prompt-injection scanning, blocked patterns and rate limits; and an asynchronous PII/PHI pass over outbound content that can allow, flag, redact, or block, producing a structured audit verdict that never carries the raw sensitive value.

The provenance ledger behind it — which model, at which temperature, from which prompt, on which substrate, produced each output — was found by the August audit to be silently recording nothing. It now **fails closed at boot** rather than reporting success: the platform refuses to start if the ledger is unreachable, on the reasoning that boot is the one moment where refusing to continue costs nothing already in flight.

> **Placement — the enterprise unlock**
>
> Every provider the gateway can reach is classified by three facts: the substrate it runs on (shared frontier API, frontier-in-your-cloud, or self-hosted), the regions it can be pinned to, and whether it carries a zero-data-retention guarantee. A tenant requiring EU residency can only be routed to a provider whose placement includes the EU; a tenant requiring zero retention only to a substrate contractually capable of it; an air-gapped tenant only to self-hosted inference. This is what makes a large pharmaceutical customer procurable, and it is architectural rather than configurational.

The RIM — accumulating judgment without a model

The Regulatory Intelligence Model sits beneath the assistant and is deliberately **not** an LLM. It observes regulatory work and aggregates signals — recurring deficiencies, reviewer triggers, data gaps, consistency issues — into durable, tenant-scoped patterns whose confidence rises with repeated observation but is bounded, so a single noisy loop can never assert certainty. It is pure and deterministic: no network calls, no wall-clock dependency, identity derived from inputs, so the same call sequence always yields the same state — a prerequisite for anything influencing a governed document.

SECTION 06

Trust architecture

In regulated life sciences, output that cannot be defended under inspection has no value. This section states where each control genuinely stands — including the two the August audit found failing.

Mechanisms, and their actual state

CONTROL	STATE	WHAT IS TRUE TODAY
Electronic signature	BUILT	Server-side credential re-verification, signing-authority and signer-role resolution, binding to a specific document version. The audit called the design right; the remaining work is consolidating to one system of record.
Clause-level lineage	BUILT	The editor cannot save content without clause-level provenance — enforced in-transaction and structurally gated in CI. A user can select any passage and ask where it came from.
Audit chain	IN FLIGHT	Chain enforcement exists on the reference store via an insert trigger. The August audit found the chain unpopulated and three integrity checks reporting an empty chain as intact. Checks now return an explicit `unverifiable` state; substrate consolidation and immutability grants are Phase 2.
Tenant isolation	IN FLIGHT	787 policies with correct USING/WITH CHECK semantics and FORCE RLS. They were *inert* because the application connected as a table-owning superuser. A least-privilege `app_service` role is now provisioned; running production under it is the remaining step.
Authentication boundary	BUILT	Default-deny gate mounted once ahead of all route registration. Algorithm-pinned JWT with CI-enforced rotation, bcrypt at 12 rounds, attempt-limited email OTP. The org-admin privilege escalation the audit exploited is closed.
Qualification pack	GENERATED	Eleven Part 11 controls mapped to the tests that exercise them, all eleven executing and passing, with a verdict that fails closed on any control lacking evidence. Execution and signature require a competent person.

The principle the audit tested

The audit's sharpest observation was not about any single control. It was that several failures shared a shape: *a control that reports success while doing nothing*. An integrity monitor verifying an all-null chain as healthy. A tenant-offboarding path reporting permanent deletion while the data survived. Governed AI commands returning past-tense success for work that never happened. In a regulated tool a silent-pass control is more dangerous than an absent one, because only the first gets relied on during an inspection.

WHAT CHANGED AS A RESULT

The remediation was not only to fix each defect but to make the class detectable: integrity checks that cannot verify now say so; the AI provenance ledger refuses boot rather than logging silently; a CI gate rejects tenant-blind data models — the class behind two of the isolation findings. Fixing an instance closes one finding. Making the shape of the mistake fail loudly closes the ones not yet written.

SECTION 07

Regulatory coverage

The submission journeys are what customers buy first. They sit inside a span running from grant funding to post-approval change — and it is the stages either side of submission, where the lifecycle suites compete, that no submission vendor has touched.

What the platform spans

Each stage below is backed by tables and services in the codebase, not roadmap.

Funding Opportunities, proposals, awards, milestones, closeout	**Preclinical** Nonclinical safety, first-in-human dose engine, M2.4 / M2.6	**Study design** 23 protocol tables, objectives through amendments	**Ethics** IRB submissions, sites, consent, reportable events
Clinical ops Risk-based monitoring — site risk, central statistical monitoring	**Supply** Suppliers, materials, batches, cold-chain readings	**Study reporting** 27 CSR tables — the deepest family in the schema	**Quality** QMS and QC — audits, OOS investigations, batch release
Submission 510(k) · CER/PER · IND/NDA — assembly, validation, transmission	**Labeling** SPL generation, prescribing information, SmPC QRD	**Post-approval** CMC change classified against FDA and EMA rules	**Post-market** E2B(R3) ICSR transport, vigilance, registrations

Inside the submission band

The chain runs intake → evidence → grounded draft → review and e-signature → eCTD or eSTAR assembly with PDF/A rendering and checksum manifest → structural and regional validation → governed transmission, each step governed by the mechanisms in Section 06.

The three submission journeys

JOURNEY	SCOPE	POSITION AGAINST 2026 TABLE STAKES
510(k) / De Novo GATED ON TEMPLATES	FDA device premarket notification. eSTAR generation, predicate identification, substantial-equivalence argumentation, filing readiness.	Substantial-equivalence engine, versioned template registry, product-code-to-standards mapping and claim-to-source traceability are live; multi-user authoring with Part 11 audit exceeds the device-side field. Official template packs are a procurement gate the system fails closed on.
CER / PER UI WIRING IN FLIGHT	EU MDR and IVDR clinical evaluation. Annex XIV structure, MEDDEV 2.7/1 rev 4 conformance, literature screening, vigilance integration.	Rule packs, conformance validator, governed export renderers and MAUDE/FAERS clients are real; screening decisions persist with an enforced exclusion rationale. Human-in-the-loop provenance is decisive — notified bodies reject pure-AI evaluations.
IND / NDA VALIDATOR LICENCE PENDING	Drug submissions via eCTD. Backbone assembly, lifecycle operations, study tagging, US Module 1, PDF/A pipeline, gateway transmission.	The spine-backed path assembles an ICH v3.2.2 backbone with MD5 manifest and genuine rendered PDF leaves; the legacy path marks leaves `rendered="false"` and names the blocker rather than emitting an untransmittable package. The reference-validator licence is procurement.

Agency transport

Thirteen gateway adapters — FDA ESG, EMA CESP, PMDA, MHRA, Health Canada, TGA eBS, Swissmedic, NMPA, MFDS, ANVISA, CDSCO Sugam, HSA PRISM — sit behind one governed transmit path with pre-transmit checks and bundle-integrity verification. Because there is no FDA certification programme for eCTD software, direct submission on sponsor-held accounts is reachable without an incumbent's permission.

SECTION 08

Chemistry, manufacturing, and controls

CMC is where a submission meets manufacturing reality. It is also where a one-time filing becomes a continuing relationship — every manufacturing change for the life of the product has to be classified, justified, and reported.

The Module 3 compiler — the provenance argument, concretely

Source objects — specifications, analytical methods, stability studies, batch records, manufacturing processes, container-closure and comparability data — compile deterministically into the CTD 3.2.S drug-substance and 3.2.P drug-product subsections. Every compiled section carries a content hash, a staleness flag with a stated reason, and **per-section lineage recording which source object produced it and that source's hash at the moment of compilation**.

That last property is the platform's central claim made specific: when a stability study or a specification changes, every Module 3 section derived from it is marked stale *and told why*, rather than silently drifting out of date until a reviewer finds the inconsistency. Elsewhere in this paper provenance is an architectural principle; here it is a compiler.

Deterministic engines, not model calls

The CMC module follows the same doctrine as the RIM — the judgments a reviewer will challenge are computed, reproducible, and citable, not generated.

SUPAC / variations classifier

A pure function — no database, no model, no side effects. A proposed manufacturing change returns an FDA reporting category (Annual Report, CBE-0, CBE-30, or prior approval), a SUPAC tier, an EMA variation class (IA / IAIN / IB / II), bioequivalence requirements, the impacted CTD sections, and citations to the controlling guidance. Grounded in SUPAC-IR/MR/SS, 21 CFR 314.70, ICH Q12, and EC No 1234/2008.

Shelf-life estimation — ICH Q1E

Ordinary least-squares fit of the stability attribute against time, solved to where the one-sided 95% confidence limit meets the specification limit, with the t-quantile evaluated against a tested Student-t distribution so the result reproduces exactly. The code states its own scope limit: single attribute, single factor; batch poolability is a separate module rather than an implied capability.

ICH compliance checker

Deterministic evaluation of a project's real stored data against Q1A(R2), Q2(R1), Q3A/Q3B, Q3D(R2), Q6A/Q6B, Q8(R2), Q9, and Q10 — the guidelines most frequently cited when FDA refuses to file.

Quality-by-design analyzer

Derives critical quality attributes from specification test parameters, impurity profiles, and stability indicators, and critical process parameters from manufacturing process records and in-process controls — from the project's actual data, replacing type-string heuristics wherever a real project exists.

WHY THIS MODULE CARRIES COMMERCIAL WEIGHT

A submission platform sells once per filing. A change-control system is used continuously for the commercial life of the product: every process change, site transfer, supplier substitution, and scale-up must be classified against both FDA and EMA rules and reported in the right category. The change-control store is org-scoped and real, and its verdicts are *computed by the classifier at read time rather than stored* — so a change in the rules re-projects every historical change rather than leaving stale conclusions in the database. That is the difference between a submission tool and a regulatory system of record.

SECTION 09

Engineering scale and the module portfolio

Scale alone is not an asset. Scale that stays modifiable, under mechanical invariant enforcement, in a domain that cannot be compressed — that is.

What exists, counted

DIMENSION	MEASURE	NOTE
TypeScript modules / lines	5,053 / 1.44M	~1.64M lines across 7,805 files counting all languages
Tables provisioned from blank	931	Single idempotent installer, verified exit 0
Row-level security policies	787	Correct USING / WITH CHECK semantics, FORCE RLS
API route handlers / modules	3,846 / 389	Behind a global default-deny boundary
Registered AnA tools	359	Across 13 capability categories
Test files / assertions passing	1,694 / 19,181	Unit, route, database, security, and golden-journey suites
CI guard scripts / workflows	79 / 19	Governance, security, and schema-integrity gates

Invariants enforced by machine

The CI guard scripts are the unusual asset. They do not check style; they check architectural invariants that would otherwise decay silently — that no code bypasses the AI gateway, that the save path cannot skip the lineage gate, that data models are not tenant-blind, that migrations are reachable, that no mock data or development authentication path reaches production, that JWT verification stays algorithm-pinned, and that the completion ledger cannot claim done without evidence.

WHERE THE QUALITY SYSTEM WAS ITSELF WEAK

The August audit's most consequential finding was about this layer: a global database mock across all test files meant the two "database-backed, safety-critical" jobs ran against a stub, request-path coverage sat at 5–13%, and no CI job ran in the RLS-enforced mode production mandates. That is why the other blockers shipped green. The mock is gone and golden-journey tests now drive all three journeys end to end, but raising coverage and making the browser tier real remain open — the audit rightly called this the highest-leverage investment, because it protects every other fix.

The domain modules behind the journeys

The three journeys are the products; these are the engines behind them, alongside CMC (Section 08) and preclinical. The August audit judged this material faithful to source — PMA against 21 CFR 814.20(b), EU CTA against CTR 536/2014, MDR Annex II, CTD Modules 2–5 — and better than most competitors' first attempts.

MODULE	SVCS	SCOPE
Global RI	48	Multi-market registration intelligence and market-specific specifications
Report engine	38	Immutable sealed report records, lineage-trace reporting, portfolio and regional renderers
IND lifecycle	36	Amendments, annual reports, briefing books, cross-references, E2B(R3) ICSR composition and gateway transport
IVD knowledge	28	Standards, legal, regulatory and scientific knowledge for diagnostics
Biostatistics	24	Group-sequential boundaries, BOIN dose finding, enrolment forecasting, external control, analytical and clinical performance
Pathway engines	22	eSTAR, PreSTAR, PMA, MDR/IVDR, CTIS, PMDA, and device-assembly manifests
Precedent intelligence	22	Approval precedent and refuse-to-file pattern libraries

SECTION 10

Independent audit and verified remediation

The platform was audited across fifteen domains on 10 August 2026 by running it, and returned not ready for general availability. Every severe mechanism it named has since been re-checked directly in the code.

14	149	19,181	132
SEVERE MECHANISMS RE-VERIFIED CLOSED	REMEDIATION COMMITS IN FOUR DAYS	TESTS PASSING ACROSS 1,694 FILES	CI ACTIONS PINNED TO A COMMIT SHA

What the audit was

Each domain was audited by an independent pass that read the code, ran commands, and queried a live database, with every severe finding routed to an adversarial cross-check whose default was to refute. The worst were reproduced by hand against a from-scratch 931-table instance with the production build booted from `dist/`. Its central observation: several controls *reported success while doing nothing*.

Finding by finding, as the code stands today

WHAT THE AUDIT FOUND	NOW	VERIFIED IN THE CURRENT CODE
eCTD compile emitted zero leaf files	CLOSED	A deterministic renderer produces genuine, byte-stable PDF leaves; the spine-backed path assembles a real ICH v3.2.2 package.
Production could not boot under enforced isolation	CLOSED	The startup posture probe runs inside a declared system tenant scope — it no longer trips the guard it should pass.
App connected as a superuser, making RLS inert	PROVISIONED	Least-privilege `app_service` role provisioned by the installer, grants refreshed each deploy. Running production under it is the remaining step.
Any signup could delete any tenant	CLOSED	The org-admin escape hatch is scoped so a platform-role guard cannot receive a stand-in; tenant management sits behind platform-admin middleware.
MFA bypassable by password alone; SSO redirect takeover	CLOSED	Pre-MFA and refresh token classes rejected at every privileged route; SSO return targets and org selection validated.
AI provenance ledger recorded nothing	CLOSED	A boot invariant fails the platform closed if the ledger is unreachable, while the per-call writer stays non-blocking.
Integrity checks called an empty chain healthy	CLOSED	An explicit `unverifiable` state with a stated reason; the Part 11 surface no longer asserts integrity it never checked.
Unencrypted database dump in the tree; DB TLS unverified	CLOSED	Dump removed and prohibited in the deploy script; TLS verification enabled and held by a regression test.
Global database mock across all tests	CLOSED	Gone from the shared setup; golden-journey tests drive all three journeys end to end, asserting honest refusals as passing.
Fabricated programme data; frontend cancelled every timer	CLOSED	Invented programme identity removed from client data; the brute-force timer optimizer removed and client error reporting wired.
Supply chain and accessibility ungoverned	CLOSED	All 132 CI action references SHA-pinned, the AGPL dependency off the ingestion path, and committed-secrets and token-contrast gates in CI.

WHAT THIS TABLE DOES AND DOES NOT ESTABLISH

Each row was confirmed by reading the current code. What has *not* been done is an independent re-scoring: the audit's headline counts were a 10 August snapshot, superseded by the work above, and no second audit has re-measured them. Treat the aggregate as history and this table as evidence. What remains genuinely open is listed in Section 12.

SECTION 11

Defensibility and moat

Four candidate advantages, sorted honestly by whether they compound, whether they are merely current, and whether they exist yet.

1 · The spanning architecture BUILT

Intake through transmission, governed end to end, across device and drug. The volume of irreducible domain code is the barrier, and it is already paid for. This is a *durable but not compounding* advantage: a well-funded competitor could cross it, but not quickly, and not without doing it twice.

2 · Governed provenance BUILT

Clause-level lineage enforced in-transaction, with a save path that structurally cannot skip it. Retrofitting this into an existing authoring product means rewriting its persistence layer — which is why the AI authoring entrants are unlikely to add it and the publishing incumbents are unlikely to add drafting.

3 · Placement-aware inference BUILT

Residency, retention, and air-gap constraints resolved per tenant at routing time. The prerequisite for large-pharma procurement. Currently differentiating, but *replicable* — expect it to become table stakes within two years as the EU AI Act phase-in makes it a standard question.

4 · Outcome data NOT BUILT

Pairs of submitted content and the agency's actual response. The only advantage here that genuinely compounds, and the only one no competitor can purchase. **Nothing captures it today.** It is designed for and it is the highest-leverage open item on the roadmap.

The compounding asset, stated precisely

Frontier model capability is available to every competitor on identical terms and improves for all of them simultaneously; no durable advantage can rest on it. What can compound is a deterministic, tenant-scoped, versioned record of what regulatory reviewers actually objected to, accumulated across customers and submissions. The RIM is the substrate designed to hold that record — and the substrate had to be trustworthy before the data was worth collecting, which is the correct ordering but means the moat is under construction rather than in place.

What is not a moat

Three things are worth naming as *not* defensible, because an investor will test them. The tool count is a capability inventory, not a barrier — a competitor with the same model access can add tools. The line count is a measure of surface area, and the audit demonstrated that surface area without coverage is a liability as much as an asset. And the design system, while genuinely good, is a matter of taste that any well-resourced team can match. The defensibility argument rests on the four items above and on the switching-cost window, not on scale.

THE HONEST ASYMMETRY

Items 1, 2 and 3 are built and real. Item 4 — the only one that compounds — is not. An investment here is a bet that the team converts a strong architectural position into the operational trustworthiness required to acquire customers, and then converts those customers into the outcome dataset that makes the position permanent. Both conversions are execution, and the audit is the best available evidence of how this team executes when told it is wrong.

SECTION 12

Roadmap to general availability

Sequenced so each phase unblocks the next and produces evidence a buyer's reviewers accept. This is a dependency order, not a wish-list, and it is the audit's own programme.

PHASE	GOAL	PRINCIPAL WORK AND ITS EVIDENCE GATE
0 LARGELY DONE	**Stop the bleeding, and make CI tell the truth** Weeks 1–2	Close the tenant-deletion path, rotate the exposed credential, remove the unencrypted dump, scope the test database mock, remove the timer optimizer, fix the SSO redirect and MFA token-class bypass. *Gate: every later fix becomes provable.* Verified closed in Section 10.
1 IN FLIGHT	**Make isolation real, and make it boot** Weeks 3–8	Run under the least-privilege role; fix the tables that would then deny-all; add residual tenant keys to route-reachable orphan tables; fix the startup posture probe so production boots under enforcement; supply full environment in every deploy target; move submission output to durable storage. *Gate: a CI job boots the built image in production mode and asserts readiness returns 200.*
2 BEGUN	**Make the regulated core trustworthy** Weeks 6–16	Land chain and immutability on the durable applier; enforce immutability with triggers and revoked grants; serialize chain writes; consolidate to one signature system of record; replace advisory grounding with retrieval-verified citation binding; pin dated model snapshots. *Gate: an independent Part 11 gap assessment against the deployed schema.*
3 NEXT	**Make one journey actually complete** Weeks 10–22	One eCTD pathway emitting a conformant sequence that passes an external validator; ship the review loop; correct filing-type to pathway mapping; expose what is built in navigation; build the privacy lifecycle. *Gate: an automated golden journey that reopens the package and validates it.*
4 BEFORE GA	**Earn the outside signatures**	Third-party penetration test; independent Part 11 gap assessment; restore-from-backup drill with documented objectives; raised coverage floors and a real browser tier; a VPAT passing WCAG 2.2 AA; close the licensing and supply-chain gaps. *Gate: evidence a buyer's security, quality, and regulatory teams accept on paper.*

Procurement, running in parallel

Independent of engineering: official eSTAR templates and field maps, eCTD document type definitions, the reference validator licence, agency gateway credentials, and a medical terminology dictionary licence. These are single-source and the platform correctly refuses to approximate them — which converts part of the timeline from an engineering problem into a commercial one and should be underwritten as such.

REALISTIC TIMELINE

A friendly, single-tenant, caveated design-partner pilot is reachable at the end of Phase 1. A defensible first paying multi-tenant pharma customer — one whose quality and security teams sign off — is the full programme, estimated at 4–7 months with a 6–9 person team. The variance sits in validation and third-party evidence, which is calendar-bound rather than effort-bound: more engineers do not shorten a penetration test or a qualification signature.

SECTION 13

Risks, and what to underwrite

What could go wrong, stated at the same standard as the rest of this document.

Execution risk on trustworthiness

The severe mechanisms an outside audit named are closed, but the regulated core is not yet through Phase 2 and no second audit has re-scored the platform. The risk is not that the work is unknown — it is scoped — but that hardening programmes routinely take longer than estimated, and this one runs on a clock set by the standards migration.

Procurement dependency

Licensed templates, validator licences, and gateway credentials are single-source. No amount of engineering conjures them, and the platform is designed to refuse rather than approximate. A delay here delays a first filing regardless of code readiness.

Incumbent response

The largest RIM vendors began shipping AI agents in 2026. Their advantage is distribution into accounts this platform must win from outside. The window in which their AI is a workflow bolt-on rather than a drafting engine is finite and not controlled by this company.

Breadth versus depth

Spanning device and drug is the differentiator and also the execution risk. The audit's judgment — make *one* journey complete before broadening — is the right sequencing, and it means near-term progress will look narrower than the platform's eventual scope.

Validation is a recurring cost

Regulated customers require vendor validation evidence, refreshed as the product changes. This is an ongoing operational burden on every release, not a one-time engineering task, and it structurally slows shipping velocity after GA.

No commercial proof yet

This paper contains no revenue, pipeline, or customer figures because none are derivable from the codebase. The technical case stands on its own; the commercial case is a separate diligence item and should be requested separately.

What an investment is buying

A spanning architecture in a market of point tools, built against a standards migration that has forced every sponsor's stack open, with the domain depth already paid for and the remaining work scoped by an independent audit rather than estimated internally. What it is not buying is a finished product: the platform is pre-GA, its regulated core is mid-hardening, and the compounding data moat is designed but not collecting.

IN SUMMARY

What has been built is the hard part: a governed chain from source evidence to transmitted submission, with provenance enforced structurally rather than by policy, across both regulatory worlds. What was wrong with it has been independently found, published here rather than buried, and is being closed at a measurable rate. The market is fragmented, the migration has forced it open, and the architecture that spans it exists in code today — the question is execution against a known list, on a clock, and that is the risk being priced.

Prepared August 2026 against the `concept2cure-v2` engineering branch. Audit figures are from an independent 15-domain readiness assessment dated 10 August 2026; remediation claims in Section 10 were re-verified in the code for this document.